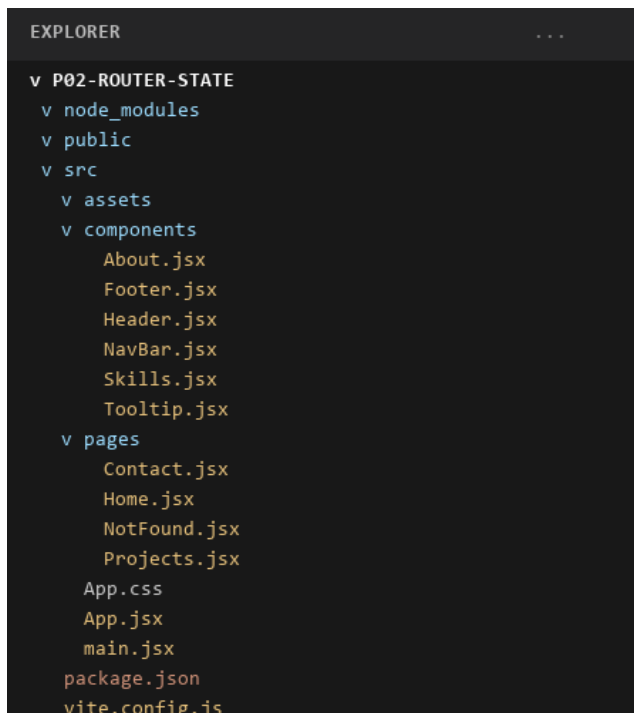


Practical 2: State Management and Routing in React

Problem Definition:

- Extend the portfolio application from Practical 1 by adding React Router.
- Implement a navigation bar with at least 3 routes: Home, Projects, and Contact.
- Each route must render a distinct component without a full page reload.
- Add at least 2 useState variables used meaningfully: one for toggling UI visibility and one for managing a form input on the Contact page.
- The Contact page must include a controlled input that captures and displays user input in real time.

Folder Structure



Implementation:

main.jsx

p02-router-state > src > main.jsx

```
1  import React from 'react';
2  import ReactDOM from 'react-dom/client';
3  import { BrowserRouter } from 'react-router-dom';
4  import App from './App.jsx';
5  import './index.css';
6
7  ReactDOM.createRoot(document.getElementById('root')).render(
8    <React.StrictMode>
9      <BrowserRouter>
10        <App />
11      </BrowserRouter>
12    </React.StrictMode>,
13  );
14
```

Ln 14, Col 1 Spaces: 2 UTF-8 JavaScript

App.jsx

p02-router-state > src > App.jsx > App

```
1  import React, { useState, useEffect } from 'react';
2  import { Routes, Route } from 'react-router-dom';
3  import NavBar from './components/NavBar';
4  import Footer from './components/Footer';
5  import Home from './pages/Home';
6  import Projects from './pages/Projects';
7  import Contact from './pages/Contact';
8  import NotFound from './pages/NotFound';
9  import './App.css';
10
11  function App() {
12    // useState: Dark/Light Mode Theme Toggle State
13    const [darkMode, setDarkMode] = useState(true);
14
15    const toggleTheme = () => {
16      setDarkMode(prev => !prev);
17    };
18
19    useEffect(() => {
20      if (!darkMode) {
21        document.body.classList.add('light-theme');
22      } else {
23        document.body.classList.remove('light-theme');
24      }
25    }, [darkMode]);
26
27    return (
28      <div className={`app-container ${!darkMode ? 'light-theme' : ''}`}>
29      // ... (truncated remaining lines)

```

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NavBar.jsx

p02-router-state > src > components > NavBar.jsx > NavBar

```
1  import React from 'react';
2  import { NavLink } from 'react-router-dom';
3  import { Code2, Home, FolderKanban, Mail, Sun, Moon } from 'lucide-react';
4
5  function NavBar({ darkMode, toggleTheme }) {
6    return (
7      <nav className="navbar" role="navigation" aria-label="Main Navigation">
8        <div className="container nav-flex">
9          <NavLink to="/" className="brand-logo">
10             <Code2 size={28} />
11             <span>Ranak<span className="gradient-text">.Dev</span></span></NavLink>
12          </NavLink>
13
14          <ul className="nav-links">
15            <li>
16              <NavLink
17                to="/"
18                end
19                className={({ isActive }) => `nav-link ${isActive ? 'active' : ''}`}
20              >
21                <Home size={18} />
22                <span>Home</span>
23              </NavLink>
24            </li>
25            // ... (truncated remaining lines)
```

Ln 61, Col 1 Spaces: 2 UTF-8 JavaScript

Home.jsx

p02-router-state > src > pages > Home.jsx > Home

```
1  import React from 'react';
2  import Header from '../components/Header';
3  import About from '../components/About';
4  import Skills from '../components/Skills';
5  import { studentInfo } from '../data/studentData';
6  import { ArrowRight, Star } from 'lucide-react';
7  import { Link } from 'react-router-dom';
8
9  function Home() {
10    const featuredProjects = studentInfo.projects.filter(p => p.featured);
11
12    return (
13      <div className="home-page">
14        <Header
15          name={studentInfo.name}
16          role={studentInfo.role}
17          themeColor="#6366f1"
18        />
19
20        <div className="container">
21          <About />
22
23          <Skills skillllist={studentInfo.skills} />
24
25          <section style={{ marginTop: '3rem', marginBottom: '2rem' }}>
26            <div style={{ display: 'flex', justifyContent: 'space-between', alignItems: 'center', marginBottom: '1.5rem' }}>
27              // ... (truncated remaining lines)
```

Ln 58, Col 1 Spaces: 2 UTF-8 JavaScript

Projects.jsx

p02-router-state > src > pages > Projects.jsx > Projects

```
1  import React, { useState } from 'react';
2  import { FolderKanban, Filter } from 'lucide-react';
3  import { studentInfo } from '../data/studentData';
4
5  const GithubIcon = () => (
6    <svg width="16" height="16" viewBox="0 0 24 24" fill="none" stroke="currentColor" strokeWidth="2" strokeLinecap="round" strokeL
7      <path d="M15 22v-4a4.8 4.8 0 0 0 1-3.5c3 0 6-2 6-5.5.08-1.25-.27-2.48-1-3.5.28-1.15.28-2.35 0-3.5 0 0-1 0-3 1.5-2.64-.5-5.36-
8      <path d="M9 18c-4.51 2-5-2-7-2"></path>
9    </svg>
10 );
11
12 function Projects() {
13   const [activeCategory, setActiveCategory] = useState('All');
14
15   const categories = ['All', 'React', 'React Router', 'Backend'];
16
17   const filteredProjects = activeCategory === 'All'
18     ? studentInfo.projects
19     : studentInfo.projects.filter(p => p.category.toLowerCase().includes(activeCategory.toLowerCase()));
20
21   return (
22     <div className="container">
23       <div className="projects-header">
24         <h1 className="section-title" style={{ fontSize: '2.5rem' }}>
25           <FolderKanban size={32} /> My Projects & Portfolio
26         </h1>
27       </div>
28     </div>
29   );
30 }
```

Ln 77, Col 1 Spaces: 2 UTF-8 JavaScript

Contact.jsx

p02-router-state > src > pages > Contact.jsx > Contact

```
1  import React, { useState } from 'react';
2  import { Mail, Send, HelpCircle, Eye, CheckCircle, MessageSquare } from 'lucide-react';
3  import Tooltip from '../components/Tooltip';
4
5  function Contact() {
6    // useState 1: Controlled Form Input States
7    const [formData, setFormData] = useState({
8      name: '',
9      email: '',
10     message: ''
11   });
12
13   // useState 2: Tooltip / Element Visibility Toggle State
14   const [showHelpTooltip, setShowHelpTooltip] = useState(false);
15
16   // useState 3: Form submission feedback state
17   const [submitted, setSubmitted] = useState(false);
18
19   const MAX_CHARS = 200;
20
21   const handleChange = (e) => {
22     const { name, value } = e.target;
23     setFormData(prev => ({
24       ...prev,
25       [name]: value
26     }));
27   };
28
29   const handleSubmit = (e) => {
30     e.preventDefault();
31     if (!formData.name || !formData.message) return;
32     setSubmitted(true);
33     // ... (truncated remaining lines)
```

Ln 174, Col 1 Spaces: 2 UTF-8 JavaScript

Tooltip.jsx

p02-router-state > src > components > Tooltip.jsx > Tooltip

```
1  import React from 'react';
2  import { HelpCircle, X, Info } from 'lucide-react';
3
4  function Tooltip({ isVisible, onClose }) {
5    if (!isVisible) return null;
6
7    return (
8      <div className="tooltip-card">
9        <div style={{ display: 'flex', justifyContent: 'space-between', alignItems: 'flex-start', marginBottom: '0.5rem' }}>
10          <h4 style={{ color: 'var(--accent-primary)', display: 'flex', alignItems: 'center', gap: '0.4rem' }}>
11            <Info size={18} /> How State Management Works
12          </h4>
13          <button
14            onClick={onClose}
15            style={{ background: 'none', border: 'none', color: 'var(--text-secondary)', cursor: 'pointer' }}
16            aria-label="Close Tooltip"
17          >
18            <X size={18} />
19          </button>
20        </div>
21        <p style={{ fontSize: '0.88rem', color: 'var(--text-secondary)', lineHeight: '1.5' }}>
22          This box toggles visibility dynamically using a <code>useState</code> boolean variable. When you type in the form inputs
23        </p>
24      </div>
25    );
26  }
27
28  // ... (truncated remaining lines)
```

Ln 29, Col 1 Spaces: 2 UTF-8 JavaScript

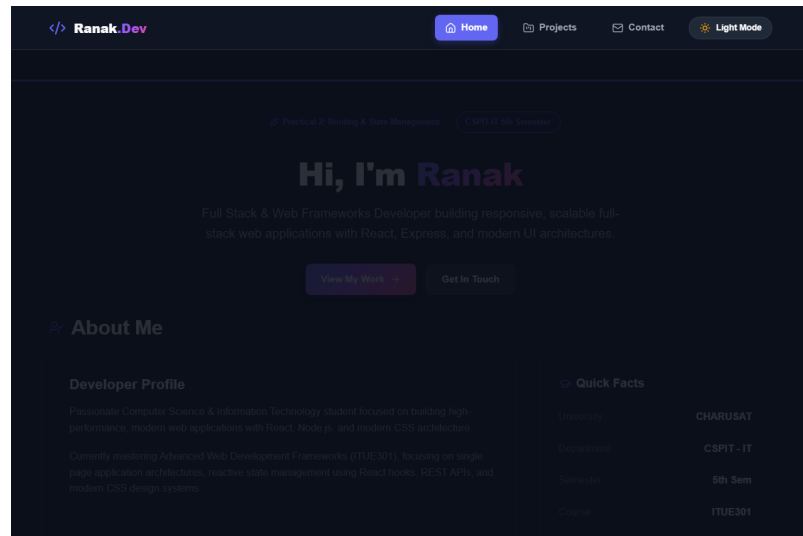
NotFound.jsx

p02-router-state > src > pages > NotFound.jsx > NotFound

```
1  import React from 'react';
2  import { Link } from 'react-router-dom';
3  import { Home, AlertTriangle } from 'lucide-react';
4
5  function NotFound() {
6    return (
7      <div className="container not-found-container">
8        <div className="glass-card" style={{ padding: '3.5rem 2rem', maxWidth: '600px', margin: '0 auto' }}>
9          <AlertTriangle size={64} color="var(--warning)" style={{ marginBottom: '1rem' }} />
10          <div className="not-found-code">404</div>
11          <h2 style={{ fontSize: '1.8rem', marginTop: '0.5rem' }}>Page Not Found</h2>
12          <p className="not-found-msg">
13            Oops! The route you are trying to access does not exist in this single page application.
14          </p>
15          <Link to="/" className="btn btn-primary">
16            <Home size={18} /> Return to Home
17          </Link>
18        </div>
19      </div>
20    );
21  }
22
23  // ... (truncated remaining lines)
```

Ln 24, Col 1 Spaces: 2 UTF-8 JavaScript

Output:



Key Questions:

- **What is the difference between client-side routing and server-side routing?**

In server-side routing, each link click sends an HTTP GET request to the server, reloading the entire HTML document. In client-side routing with React Router, JavaScript intercepts navigation, updates the browser URL via HTML5 History API, and renders the matched route component dynamically without refreshing the page.

- **Why do we use <Link> instead of <a> in React Router?**

 triggers standard browser reload, resetting all component state. <Link to="#"> prevents page refresh using e.preventDefault() and swaps components locally in the Virtual DOM.

- What is a controlled component and how does useState manage it?

A controlled component ties its input value directly to React state (value={state}) and updates that state on every keystroke through onChange={(e) => setState(e.target.value)}, keeping UI and state perfectly synchronized.

Supplementary Problems:

- Add a 404 Not Found route that renders a custom error component for unknown paths.

Implementation:

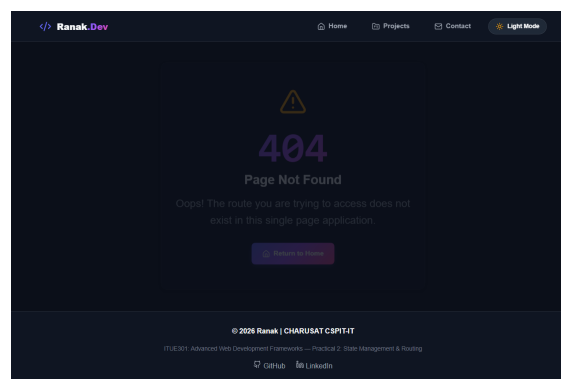
NotFound.jsx

p02-router-state > src > pages > NotFound.jsx > NotFound

```
1  import React from 'react';
2  import { Link } from 'react-router-dom';
3  import { Home, AlertTriangle } from 'lucide-react';
4
5  function NotFound() {
6    return (
7      <div className="container not-found-container">
8        <div className="glass-card" style={{ padding: '3.5rem 2rem', maxWidth: '600px', margin: '0 auto' }}>
9          <AlertTriangle size={64} color="var(--warning)" style={{ marginBottom: '1rem' }} />
10         <div className="not-found-code">404</div>
11         <h2 style={{ fontSize: '1.8rem', marginTop: '0.5rem' }}>Page Not Found</h2>
12         <p className="not-found-msg">
13           Oops! The route you are trying to access does not exist in this single page application.
14         </p>
15         <Link to="/" className="btn btn-primary">
16           <Home size={18} /> Return to Home
17         </Link>
18       </div>
19     </div>
20   );
21   // ... (truncated remaining lines)
```

Ln 24, Col 1 Spaces: 2 UTF-8 JavaScript

Output:



- Implement a dark/light mode toggle using useState that applies a CSS class to the root element.

Implementation:

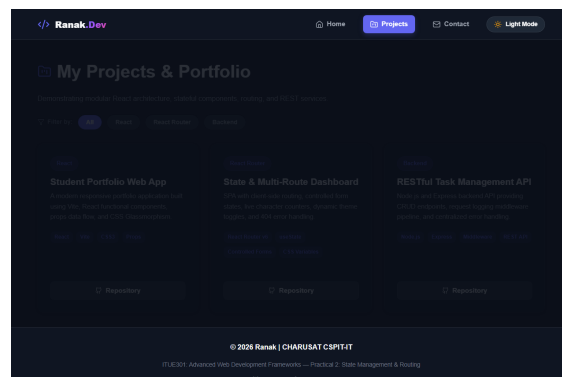
Tooltip.jsx / Home.jsx

```
p02-router-state > src > components > Tooltip.jsx > Tooltip

1  import React from 'react';
2  import { HelpCircle, X, Info } from 'lucide-react';
3
4  function Tooltip({ isVisible, onClose }) {
5    if (!isVisible) return null;
6
7    return (
8      <div className="tooltip-card">
9        <div style={{ display: 'flex', justifyContent: 'space-between', alignItems: 'flex-start', marginBottom: '0.5rem' }}>
10         <h4 style={{ color: 'var(--accent-primary)', display: 'flex', alignItems: 'center', gap: '0.4rem' }}>
11           <Info size={18} /> How State Management Works
12         </h4>
13         <button
14           onClick={onClose}
15           style={{ background: 'none', border: 'none', color: 'var(--text-secondary)', cursor: 'pointer' }}
16           aria-label="Close Tooltip"
17         >
18           <X size={18} />
19         </button>
20       </div>
21       <p style={{ fontSize: '0.88rem', color: 'var(--text-secondary)', lineHeight: '1.5' }}>
22         This box toggles visibility dynamically using a <code>useState</code> boolean variable. When you type in the form inputs
23       </p>
24     </div>
25   );
26 }
```

Ln 29, Col 1 Spaces: 2 UTF-8 JavaScript

Output:



- Store the contact form input in state and display a live character count below the input.

Implementation:

Contact.jsx

p02-router-state > src > pages > Contact.jsx > Contact

```
1  import React, { useState } from 'react';
2  import { Mail, Send, HelpCircle, Eye, CheckCircle, MessageSquare } from 'lucide-react';
3  import Tooltip from '../components/Tooltip';
4
5  function Contact() {
6    // useState 1: Controlled Form Input States
7    const [formData, setFormData] = useState({
8      name: '',
9      email: '',
10     message: ''
11   });
12
13   // useState 2: Tooltip / Element Visibility Toggle State
14   const [showHelpTooltip, setShowHelpTooltip] = useState(false);
15
16   // useState 3: Form submission feedback state
17   const [submitted, setSubmitted] = useState(false);
18
19   const MAX_CHARS = 200;
20
21   const handleChange = (e) => {
22     const { name, value } = e.target;
23     setFormData(prev => ({
24       ...prev,
25       [name]: value
26     }));
27   };
28
29   const handleSubmit = (e) => {
30     e.preventDefault();
31     if (!formData.name || !formData.message) return;
32     setSubmitted(true);
33     // ... (truncated remaining lines)
```

Ln 174, Col 1 Spaces: 2 UTF-8 JavaScript

Output:

The screenshot shows a web application interface with a dark theme. At the top, there's a navigation bar with 'Ranak.Dev', 'Home', 'Projects', 'Contact' (highlighted), and 'Light Mode'. Below the navigation bar is a section titled 'Contact & Live State Form' with a subtitle 'Demonstrating controlled inputs, live state updates, and element visibility toggle using useState'. There are two columns of content. The left column, titled 'Controlled Contact Form', contains three input fields: 'Your Name *', 'Email Address', and 'Message *'. The right column, titled 'Live State Data', displays the current state of the form: 'Name: [Your Name]', 'Email: [Email Address]', and 'Message: [Message *]'. At the bottom of the left column is a 'Send Message' button. A 'How this State Works?' link is located in the top right corner of the main content area.

Github Link : <https://github.com/Ranakeyur-31525/awdf>